THE IDEAL OF DENOMINATORS OF A REGULAR MEROMORPHIC SECTION: TWO SHORT EXACT SEQUENCES WITH SUPPORT CONDITION AND A CYCLE-THEORETIC DIFFERENCE FORMULA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02SW (sources/stacks-chow/pinned-original-source.tex line 18649).

Allowed standard prerequisites (editorial).

- The ideal of denominators of a regular meromorphic section and the associated injections. Divisors, Definition labelled divisors-definition-regular-meromorphic-ideal-denominators (Stacks tag 02P1) and Lemma labelled divisors-lemma-make-maps-regular-section (Stacks tag 02P2), invoked at statement line 18667 and proof lines 18715-18719. Supply the ideal sheaf I of denominators and the injective maps 1 : I F -> F and s : I F -> F tensor L whose cokernels are supported on a nowhere dense closed subset; the identification of the quotient sheaves and the cycle computation are inside the counted span.
- Pullback of meromorphic sections along the components and regularity of the restrictions. Divisors, Lemmas labelled divisors-lemma-pullback-meromorphic-sections-defined (Stacks tag 02OU) and divisors-lemma-meromorphic-sections-pullback (Stacks tag 02OY), invoked at proof lines 18722-18724. Supply that the restrictions s_i to the components X_i of delta-dimension k+1 are defined and regular meromorphic for L|_{X_i}.
- The module-level comparison of the two lengths with the divisor sum. Same chapter, Lemma labelled lemma-no-embedded-points-modules (source line 18594), invoked at proof line 18752. Supplies the numerical identity that makes the coefficient of [Z] on the two sides agree; the reduction to it, including the rewriting s = (a/b) s_xi and the identification of the coefficient formulas, is carried out in the counted span.
- The base situation (S, delta) for operational Chow groups and cycle classes. Same chapter, Situation labelled situation-setup (source line 1372), invoked

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in the statement. Fixes the delta-dimension conventions, the cycle group Z_k(X) and the pullback and cap operations used in the statement.

Source reference. Original definition 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 02P1: sources/stacks-chow/tags.txt, source lines 3536–3536. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02P2: sources/stacks-chow/tags.txt, source lines 3537–3537. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 18715 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02OU: sources/stacks-chow/tags.txt, source lines 3529–3529. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 18722 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02OY: sources/stacks-chow/tags.txt, source lines 3533–3533. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 18723 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02QH: sources/stacks-chow/pinned-original-source.tex, source lines 18594–18594. Same-chapter pointer into the author environment lemma at source line 18594; invoked at proof line 18752 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Let $s \in \Gamma(X, \mathcal{K}_X(\mathcal{L}))$ be a meromorphic section of $\mathcal{L}$. Assume*

1. $\dim_\delta(X) \leq k + 1$,
2. *X has no embedded points,*
3. *$\mathcal{F}$ has no embedded associated points,*
4. *the support of $\mathcal{F}$ is X, and*
5. *the section s is regular meromorphic.*

In this situation let $\mathcal{I} \subset \mathcal{O}_X$ be the ideal of denominators of s, see Divisors, Definition 02P1. Then we have the following:

1. *there are short exact sequences*

$$\begin{array}{ccccccccc} 0 & \to & \mathcal{I}\mathcal{F} & \xrightarrow{1} & \mathcal{F} & \to & \mathcal{Q}_1 & \to & 0 \\ 0 & \to & \mathcal{I}\mathcal{F} & \xrightarrow{s} & \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L} & \to & \mathcal{Q}_2 & \to & 0 \end{array}$$

2. *the coherent sheaves $\mathcal{Q}_1$, $\mathcal{Q}_2$ are supported in δ-dimension $\leq k$,*
3. *the section s restricts to a regular meromorphic section s_i on every irreducible component X_i of X of δ-dimension $k + 1$, and*

(4) *writing* $[\mathcal{F}]_{k+1} = \sum m_i[X_i]$ *we have*

$$[\mathcal{Q}_2]_k - [\mathcal{Q}_1]_k = \sum m_i (X_i \to X)_* div_{\mathcal{L}|_{X_i}}(s_i)$$

in $Z_k(X)$*, in particular*

$$[\mathcal{Q}_2]_k - [\mathcal{Q}_1]_k = c_1(\mathcal{L}) \cap [\mathcal{F}]_{k+1}$$

in $\mathrm{CH}_k(X)$.

Proof. Recall from Divisors, Lemma 02P2 the existence of injective maps $1 : \mathcal{IF} \to \mathcal{F}$ and $s : \mathcal{IF} \to \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}$ whose cokernels are supported on a closed nowhere dense subsets T. Denote $\mathcal{Q}_i$ there cokernels as in the lemma. We conclude that $\dim_\delta(\mathrm{Supp}(\mathcal{Q}_i)) \leq k$. By Divisors, Lemmas 02OU and 02OY the pullbacks s_i are defined and are regular meromorphic sections for $\mathcal{L}|_{X_i}$. The equality of cycles in (4) implies the equality of cycle classes in (4). Hence the only remaining thing to show is that

$$[\mathcal{Q}_2]_k - [\mathcal{Q}_1]_k = \sum m_i (X_i \to X)_* \mathrm{div}_{\mathcal{L}|_{X_i}}(s_i)$$

holds in $Z_k(X)$. To see this, let $Z \subset X$ be an integral closed subscheme of δ-dimension k. Let $\xi \in Z$ be the generic point. Let $A = \mathcal{O}_{X,\xi}$ and $M = \mathcal{F}_\xi$. Moreover, choose a generator $s_\xi \in \mathcal{L}_\xi$. Then we can write $s = (a/b)s_\xi$ where $a, b \in A$ are nonzerodivisors. In this case $I = \mathcal{I}_\xi = \{x \in A \mid x(a/b) \in A\}$. In this case the coefficient of $[Z]$ in the left hand side is

$$\mathrm{length}_A(M/(a/b)IM) - \mathrm{length}_A(M/IM)$$

and the coefficient of $[Z]$ in the right hand side is

$$\sum \mathrm{length}_{A_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}) \mathrm{ord}_{A/\mathfrak{q}_i}(a/b)$$

where $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$ are the minimal primes of the 1-dimensional local ring A. Hence the result follows from Lemma 02QH. □